

Code variables: the geometry inputs

Code variable	Physical meaning
<code>reacty</code>	Laboratory vertical reaction position, read from <code>H.react.y</code> . Positive upward.
<code>xmis</code>	HMS vertical mispointing translation. The code places the vertex at $-\text{reacty} - \text{xmis}$ in transport x , positive downward.
<code>zfoil</code>	Assigned foil position along the laboratory beam direction.
<code>c</code>	$\cos(\text{angle})$, with <code>angle</code> the spectrometer angle in radians. The code uses <code>zfoil c</code> as the longitudinal vertex position.
<code>L</code>	Distance along the spectrometer axis from its target reference plane to the sieve plane. HMS: 168 cm.
<code>xsT</code>	Nominal transport- x coordinate of the assigned sieve-hole center. Fixed by the hole grid.

All positions and distances here are in cm. The target reference plane is at spectrometer $z = 0$.

Code variables: the constructed fit targets

Code variable	Physical meaning
<code>xptarT</code>	Desired transport-x slope for the optics fit, dx/dz . Dimensionless, usually reported in mrad.
<code>xtarT</code>	Corresponding transport-x intercept at the target reference plane, in cm. It need not equal the position at the foil.

These are geometry-derived target values supplied to the fit.

Exact code correspondence. In `targetTruth`, the input `xs` is the nominal hole coordinate stored as `xsT`. The outputs `t.xptar` and `t.xtar` become `xptarT` and `xtarT` in the fit tree.

The equations below consistently use `xsT`, `xptarT`, and `xtarT`.

The condition to be satisfied

Show that: the constructed slope and intercept describe a straight ray through the assigned sieve-hole center.

Starting at the target reference plane, propagation to the sieve adds distance times slope:

$$\underbrace{x_{\text{tarT}}}_{\text{starting position}} + \underbrace{L x_{\text{ptarT}}}_{\text{change in position}} = \underbrace{x_{\text{sT}}}_{\text{hole center}} .$$

Thus the required identity is

$$x_{\text{tarT}} + L x_{\text{ptarT}} - x_{\text{sT}} = 0.$$

All three positions must use the same transport coordinate origin. This is the reference ray to the nominal center, before considering finite hole size or scattering.

The HMS equations implemented in the code

The HMS branch of `targetTruth` constructs

$$x_{\text{ptarT}} = \frac{x_{\text{ST}}}{L - z_{\text{foil}} c},$$

$$x_{\text{tarT}} = -\text{reacty} - x_{\text{mis}} - x_{\text{ptarT}} z_{\text{foil}} c.$$

The slope uses the hole position divided by the foil-to-sieve distance.

The intercept also includes the displaced vertex, $-\text{reacty} - x_{\text{mis}}$.

Source: `spectrometer_config.h`, HMS branch of `targetTruth`.

Storage mapping: `make_fit_ntuple_from_gmm.C`.

Direct substitution exposes the contradiction

Substitute the implemented intercept, then the implemented slope:

$$\begin{aligned} & x_{tarT} + L x_{ptarT} - x_{sT} \\ &= -reacty - x_{mis} - x_{ptarT} z_{foilc} + L x_{ptarT} - x_{sT} \\ &= -reacty - x_{mis} + x_{ptarT}(L - z_{foilc}) - x_{sT} \\ &= -reacty - x_{mis} + x_{sT} - x_{sT} \\ &= \boxed{-reacty - x_{mis}.} \end{aligned}$$

The required result is zero. The implemented result is generally nonzero.

The pair reaches $x_{sT} - reacty - x_{mis}$, rather than x_{sT} . Equality holds only when $reacty + x_{mis} = 0$.

What the contradiction means

For the same longitudinal approximation, satisfying the identity requires

$$x_{\text{ptarT}} = \frac{x_{\text{sT}} + \text{reacty} + x_{\text{mis}}}{L - z_{\text{foil c}}}.$$

Use that same x_{ptarT} in the intercept equation.

Conceptually: the numerator must be the displacement from the vertex to the hole. The present numerator includes only the hole position, although the intercept retains the vertex displacement.

Scope. This establishes the inconsistency of the constructed target pair. Its effect on fitted or replayed resolution requires a separate data comparison.

Replay context confirmed by Christine. The 2024 replay used the same matrix as NPS replay, checked with the afterburner closure test. All zero-order offsets were removed for that replay. The HCANA version remains unchanged. These facts do not change the substitution above.